PETRA ZLATARIĆ-NOVOSEL

Machine Learning Engineer | NLP & Document AI

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PROFESSIONAL SUMMARY

Machine learning engineer with 5+ years of experience in NLP, document classification and predictive modeling, including deploying and operating ML systems in production. Experienced in taking language-model and classification work from dataset preparation and experimentation through model serving and production monitoring. Currently pursuing a PhD focused on named entity recognition in Croatian legal texts. Hands-on experience with Python, PyTorch, Hugging Face Transformers, MLflow and Azure.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | Slavonica Document AI d.o.o. | Osijek *03/2023–Present*

- Developed and deployed a document classifier for scanned public-administration documents, achieving 94% accuracy and 0.91 macro F1 across 23 classes; the service processes approximately 15,000 documents per day.
- Fine-tuned XLM-R and BERTić transformer models for document classification and named entity recognition in Croatian administrative texts.
- Served models as FastAPI services in Docker on Azure App Service, using Azure ML and MLflow for experimentation tracking and model registry.
- Built production monitoring with weekly reports on input text length, vocabulary and predicted class distribution; an alert on PSI above 0.2 helped identify an OCR change that reduced accuracy by 9 percentage points, which was resolved within a week.
- Built an annotated dataset of 4,000 Croatian administrative documents with annotation guidelines and two annotators, achieving inter-annotator agreement of $\kappa = 0.87$.
- Mentor two students during internships.

Data Scientist | Baranjska kreditna banka d.d. | Zagreb (remote from Osijek)
09/2021–02/2023

- Developed customer churn prediction models using gradient boosting and LightGBM for the sales team.
- Built a customer complaint topic-classification solution, progressing from a TF-IDF and logistic-regression baseline to fastText; the 11-topic classification supported weekly reporting to customer-service management.
- Produced SQL reports from an Oracle data warehouse.

Research Project Assistant | Faculty of Applied Informatics, University of Drava
10/2020–08/2021

- Created a corpus of Croatian legal texts and annotated documents; wrote supporting Python scripts.

SELECTED PROJECT

Retrieval-Augmented Question Answering Demo | Personal project *2025*

- Built a weekend prototype for question answering over Croatian laws using a sentence-embedding model, FAISS and an open-weight LLM accessed through an API.
- Prototype only; checked manually and not used in production.

EDUCATION

PhD in Computer Science | University of Drava | 2022–Present

Research topic: Named entity recognition in Croatian legal texts. Expected completion: 2027.

MSc in Computer Science (Mag. ing. comp.) | University of Drava | 2020

TECHNICAL SKILLS

Programming and data: Python, SQL, pandas, NumPy, Oracle SQL
Machine learning and NLP: PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, fastText, spaCy, sentence-transformers, FAISS
Production and MLOps: MLflow, FastAPI, Docker, Azure ML, Azure App Service, Azure Functions, Evidently, PostgreSQL, Git, Linux
Annotation: Label Studio

CERTIFICATION

Microsoft Certified: Azure AI Engineer Associate | 2024

TEACHING

Guest lecturer, NLP Summer School | 2025 | 90-minute lecture

PUBLICATION

Zlatarić-Novosel, P. et al. (2024). "Named Entity Recognition for Croatian Administrative Documents." Workshop on Slavic Language Processing.

LANGUAGES

Croatian (native) | English (C1) | German (B2)